

CO5-AT3-Level Exam3

Question-1

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a header with the SIMATS Engineering logo, a 'Secure Exit' button, the lab name 'SIMATS Online Programming Lab', and the subject 'JAVA PROGRAMMING'. A user profile for 'VANGIMALLA BALA SIVA PRASAD REDDY' is visible on the right. The main content area displays a question titled '(H) Number Palindrome' under the 'LEVEL 3 — HARD' category. The question asks to develop a code to reverse and add a number until it becomes a palindrome. It provides sample inputs and outputs, including a test case: 1. 8765, 2. -8765, 3. 0, 4. EIGHT FIVE, 5. 87.57. Below the question, there are fields for 'Your Code', 'Program Input (stdin)', and 'Output'.

Code:

```
import java.util.*;

class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        long n = sc.nextLong();

        boolean negative = n < 0;

        if (negative)

            n = -n;

        while (true) {

            long temp = n;

            long rev = 0;

            while (temp > 0) {

                rev = rev * 10 + temp % 10;
```

```

        temp = temp / 10;
    }
    if (n == rev) {
        if (negative)
            System.out.println("-" + n);
        else
            System.out.println(n);
        break;
    }
    n = n + rev;
}
}
}

```

Question-2



Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING


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QUESTIONS

LEVEL 3 — HARD

Q1
Numbers

Q2
Control

0 / 2 saved

Level 3 — Hard (H) Steps — Control

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase: 1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

Your Code

1

Program Input (stdin)

Enter input if required...

Output

Code:

```
import java.util.*;

class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int steps = 0;

        while (n != 0) {
            if (n % 2 == 0)
                n = n / 2;
            else
                n = n - 1;

            steps++;
        }

        System.out.println(steps);
    }
}
```